

Adversarial Debate Argument Generator

DTS407TC A2 Group

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Problem

- ▶ Debate preparation needs arguments that survive counterarguments.
- ▶ Single-pass LLM generation is fluent but often shallow.
- ▶ Goal: generate three stronger arguments for a chosen side.

System

- ▶ Local open-source model: Ollama qwen3.5:4B.
- ▶ Search: Tavily API with local caching.
- ▶ Backend: FastAPI.
- ▶ Frontend: React bilingual interface.

Two Strategies

Strategy	Process	Expected Strength
Single Agent	Search then generate	Fast baseline
4 Agent Debate	Constructives, rebuttals, synthesis	More robust arguments

Adversarial Flow

1. Pro Logic and Pro Evidence build support.
2. Con Logic and Con Evidence attack the motion.
3. Both sides rebut.
4. Neutral synthesis extracts the target side's best three arguments.

Evaluation

- ▶ Automatic metrics: response time, token estimate, source count, argument count, diversity.
- ▶ Human evaluation: persuasiveness, relevance, evidence quality, rebuttal awareness.
- ▶ Expected result: adversarial strategy is slower but stronger.

Live Demo

- ▶ English: AI systems should have legal personhood.
- ▶ Chinese UI demo: the same topic selector can switch to Chinese motions.
- ▶ Show both outputs, sources, metrics, and transcript.

Ethics

- ▶ Risks: one-sided persuasion, biased sources, misinformation, privacy of search queries.
- ▶ Mitigations: show sources, keep internal opposing arguments, support human review, local model.

Q&A

- ▶ Why not fine-tune?
- ▶ How is the comparison fair?
- ▶ When is the single-agent strategy better?
- ▶ What are the limits of search-based evidence?